

A Critical Evaluation of Current Methods for Exercise Prescription in Women and Men

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ABSTRACT

IANNETTA, D., E. C. INGLIS, A. T. MATTU, F. Y. FONTANA, S. POGGIAGHI, D. A. KEIR, and J. M. MURIAS. A Critical Evaluation of Current Methods for Exercise Prescription in Women and Men. *Med. Sci. Sports Exerc.*, Vol. 52, No. 2, pp. 466–473, 2020. Common methods to prescribe exercise intensity are based on fixed percentages of maximum rate of oxygen uptake ($\dot{V}O_{2\max}$), peak work rate (WR_{peak}), maximal HR ($HR_{\max}$). However, it is unknown how these methods compare to the current models to partition the exercise intensity spectrum. **Purpose:** Thus, the aim of this study was to compare contemporary gold-standard approaches for exercise prescription based on fixed percentages of maximum values to the well-established, but underutilized, “domain” schema of exercise intensity. **Methods:** One hundred individuals participated in the study (women, 46; men, 54). A cardiopulmonary ramp-incremental test was performed to assess $\dot{V}O_{2\max}$, WR_{peak} , $HR_{\max}$, and the lactate threshold (LT), and submaximal constant-work rate trials of 30-min duration to determine the maximal lactate steady-state (MLSS). The LT and MLSS were used to partition the intensity spectrum for each individual in three domains of intensity: moderate, heavy, and severe. **Results:** $\dot{V}O_{2\max}$ in women and men was $3.06 \pm 0.41 \text{ L} \cdot \text{min}^{-1}$ and $4.10 \pm 0.56 \text{ L} \cdot \text{min}^{-1}$, respectively. Lactate threshold and MLSS occurred at a greater % $\dot{V}O_{2\max}$ and % $HR_{\max}$ in women compared with men ($P < 0.05$). The large ranges in both sexes at which LT and MLSS occurred on the basis of % $\dot{V}O_{2\max}$ (LT, 45%–74%; MLSS, 69%–96%), % WR_{peak} (LT, 23%–57%; MLSS, 44%–71%), and % $HR_{\max}$ (LT, 60%–90%; MLSS, 75%–97%) elicited large variability in the number of individuals distributed in each domain at the fixed-percentages examined. **Conclusions:** Contemporary gold-standard methods for exercise prescription based on fixed-percentages of maximum values conform poorly to exercise intensity domains and thus do not adequately control the metabolic stimulus. **Key Words:** EXERCISE-INTENSITY DOMAINS, FIXED-PERCENTAGE APPROACH, EXERCISE PRESCRIPTION

Exercise intensity is the cornerstone of exercise physiology providing a guiding framework by which we perform aerobic testing, interpret physiological responses, prescribe exercise, and induce desired adaptations to training (1). However, although it is widely accepted that exercise training is a cost-effective primary intervention to maintain health and to attenuate or reverse a variety of chronic conditions (2), the optimal exercise intensity prescription needed to produce a desired effect remains elusive. The contemporary gold standard approach for assignment of exercise intensity involves normalization based on percentages of maximal rate of O_2 uptake ($\dot{V}O_{2\max}$) as identified with cardiopulmonary incremental exercise testing. For practical reasons, alternative methods commonly used involve the identification of other

maximal physiological outcomes, such as peak work rate (WR_{peak}), and maximal HR ($HR_{\max}$) that can be linked to a given percent $\dot{V}O_{2\max}$. These methods assume that: (i) exercise intensity exists upon a continuum; and (ii) the characteristics of the metabolic responses and the tolerable duration of exercise among individuals exercising at a common percentage of maximum are identical. However, the ability of the aerobic system to maintain a given work rate cannot be ascertained from an incremental-derived maximal measure (3,4). Rather, it is the dynamic responsiveness of $\dot{V}O_2$ (or its kinetics) and the achievement (or not) of $\dot{V}O_2$ steady-state that predict exercise tolerance and unmask the underlying physiological and metabolic stimulus or “intensity” (4).

In fact, the $\dot{V}O_2$ kinetics response to step-increases in exercise work rate changes in relation to distinct metabolic boundaries: the lactate threshold (LT) and the critical power (CP), which is closely related to the maximal lactate steady-state (MLSS) (4,5). These thresholds, therefore, demarcate three “domains” of exercise intensity (6). Within the moderate-intensity domain (i.e., below LT), $\dot{V}O_2$ rapidly attains steady-state within 2 to 3 min, with an efficiency ($\Delta\dot{V}O_2/\Delta\text{work rate}$) that is highly predictable (7). Exercise within this domain can be sustained for a very long period (~4 h) and results in no, or modest, accumulation of metabolites and fatigue (8), unless protracted until “exhaustion” (9). Within the heavy-intensity

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domain (i.e., between LT and CP/MLSS), the emergence of the $\dot{V}O_2$ slow component, which reflects a reduction in efficiency (i.e., greater $\Delta\dot{V}O_2/\Delta$ work rate), delays the attainment of $\dot{V}O_2$ steady state by as much as 15 min (10). Here, the greater metabolic perturbations (e.g., elevation of blood [lactate], muscle [inorganic phosphate], H^+) although stabilized over time, are associated with greater accumulation of fatigue and reduction of exercise tolerance relative to moderate-intensity exercise (8,9,11,12). Within the severe-intensity domain (i.e., above CP and MLSS), the rate of ATP resynthesis continues to rise over time (13), time to exhaustion decreases hyperbolically as intensity increases, and end-exercise typically coincides with high levels of metabolites, fatigue accumulation, and attainment (or near attainment) of $\dot{V}O_{2max}$ (9,14,15).

Between individuals, LT and the CP/MLSS can occur at different percentages of $\dot{V}O_{2max}$ (16), thus the ranges of % $\dot{V}O_{2max}$, % WR_{peak} , or % HR_{max} that define each intensity domain can vary from person to person. Therefore, selection of constant-work rate exercise based on a fixed % $\dot{V}O_{2max}$, % WR_{peak} , or % HR_{max} may not guarantee an accurate control of the exercise intensity (3,17–19). Nevertheless, this approach remains as the most commonly used in both clinical and research trials (20–23) and is recommended by current exercise prescription guidelines (24). Surprisingly, how each exercise intensity domain is distributed in terms of % $\dot{V}O_{2max}$, % WR_{peak} , and % HR_{max} within a large group of individuals, and how sex-related differences may affect these distributions remains unknown.

Given the individual variability in the relative position of LT and MLSS, it is possible that no fixed % $\dot{V}O_{2max}$, % WR_{peak} , or % HR_{max} could guarantee a uniform domain-specific allocation of individuals, with this having important implications for exercise intensity prescriptions. Thus, using a dataset of one hundred individuals for whom both LT and MLSS were directly measured and verified, the purpose of this study was to quantify the distribution of each intensity domain at discrete fixed % $\dot{V}O_{2max}$, % WR_{peak} , or % HR_{max} values.

METHODS

Participants

One-hundred healthy adults (46 women and 54 men) participated in this study. Their physical characteristics are displayed in Table 1. Data from 40 participants have been presented in

TABLE 1. Participants' physical characteristics and physiological performance variables measured during the cardiopulmonary incremental exercise test.

	All (N = 100)	Women (n = 46)	Men (n = 54)
Age (yr)	28 ± 7	26 ± 5	30 ± 8 ^a
Height (cm)	174 ± 8	169 ± 8	177 ± 7 ^a
Weight (kg)	72 ± 10	67 ± 9	76 ± 9 ^a
$\dot{V}O_{2max}$ (L·min ⁻¹)	3.62 ± 0.72	3.06 ± 0.41	4.10 ± 0.56 ^a
$\dot{V}O_{2max}$ (mL·kg ⁻¹ ·min ⁻¹)	50.7 ± 8.6	46.4 ± 6.1	54.4 ± 8.8 ^a
WR_{peak} (W)	343 ± 66	292 ± 37	387 ± 53 ^a
WR_{peak} (W·kg ⁻¹)	4.8 ± 0.8	4.4 ± 0.6	5.1 ± 0.8 ^a
HR_{max} (bpm)	184 ± 9	184 ± 9	183 ± 9

Data are presented as mean ± SD.

^aSignificance between men and women.

previous work from our group (11,25). The level of fitness of the participants spanned from untrained to well-trained, according to current classification guidelines (26). Within ten days, all participants completed the following exercise tests on a cycle ergometer: i) a cardiopulmonary ramp-incremental exercise test to exhaustion; ii) two to four 30-min constant-work rate rides for determination of MLSS. This study was approved by the local ethics review board in compliance with the latest version of the Declaration of Helsinki. Informed consent was obtained from all individual participants included in the study.

Data Collection

All testing was performed on an electromagnetically braked cycle ergometer (Velotron, RacerMate, Seattle, WA). A metabolic cart (Quark, CPET, Cosmed, Rome, Italy) was used to measure ventilatory and gas exchange variables on a breath-by-breath basis. The system was comprised of a low-dead-space turbine for the assessment of inspired and expired volumes and gas analyzers for the assessment of fractional concentrations of respired O_2 and CO_2 which were calibrated according to the manufacturers recommendation. Blood lactate concentration ($[La^-]_b$) from a finger pin-prick was assessed with a portable analyzer (Lactate Scout, SensLab GmbH, Leipzig, Germany). HR was continuously recorded during each testing session using radiotelemetry (Garmin, Olathe, KA). Participants performed the tests on separate days, with a minimum of 48 h and a maximum of 72-h interval between sessions. All testing sessions were performed at the same time of the day (±30 min). Participants self-selected their preferred cadence (in a range between 75 and 95 rpm) during their first laboratory visit. This cadence was maintained for subsequent visits. During all exercise tests, participants were blinded to the work rate and elapsed time but received visual feedback on their cadence.

Cardiopulmonary Ramp-Incremental Exercise Test

The cardiopulmonary ramp-incremental exercise test was performed to the limit of volitional exhaustion to determine $\dot{V}O_{2max}$, LT, WR_{peak} , and HR_{max} . A 4-min period of cycling at 50 W preceded the ramp-incremental portion of the test, which consisted of 25 W·min⁻¹ (1 W every 2.4 s) and 30 W·min⁻¹ (1 W every 2 s) increments for women and men, respectively. The ramp-incremental test was terminated if participants' cadence dropped more than 10 rpm below their self-selected cadence for more than 10 s, or at volitional exhaustion, despite strong verbal encouragement.

MLSS Determination

Participants performed two to four 30-min constant-work rate tests for the determination of the work rate at MLSS. MLSS was defined as the highest work rate at which the difference in $[La^-]_b$ is less than (or equal to) 1 mM between the 10th and 30th min (27). Each 30-min constant-work rate test was preceded by a 4-min baseline of 50-W to 80-W cycling, after which the work rate was instantaneously increased to a

predetermined value. This initial work rate value was identified from a mathematical model that predicts the work rate at MLSS with a high degree of accuracy (25). Depending on $[La^-]_b$ responses, the work rate for the subsequent constant-work rate trial was either decreased or increased by 10 W until the highest work rate with stable $[La^-]_b$ was established (28). These constant-work rate trials were repeated as long as the criterion for MLSS was satisfied. $[La^-]_b$ was measured during baseline and every 5 min after the work rate was increased.

Data Analyses

Ventilatory and gas exchange variables. The raw ventilatory and gas exchange data for each participant were examined by two expert physiologists to identify the $\dot{V}O_2$ corresponding to LT, as previously described (29). For each incremental and constant-work rate trial, the breath-by-breath data were edited and aberrant data lying three SD from the local mean were deleted. Thereafter, the $\dot{V}O_2$ data were interpolated on a second-by-second basis. $\dot{V}O_{2max}$ corresponded to the highest $\dot{V}O_2$ value during the ramp test computed from a 30-s rolling average. To account for the ramp- $\dot{V}O_2$ mean response time (MRT), the “back extrapolation” approach was used (30). The work rate at LT was identified after left shifting the ramp- $\dot{V}O_2$ data by the MRT. The $\dot{V}O_2$ associated with MLSS was calculated as the average of the last 5 min of the 30-min constant work rate exercise.

HR responses. The HR values associated with LT were derived by averaging 20 data points (bpm) around the corresponding $\dot{V}O_2$. The HR response during MLSS trial was calculated as the average of the last 20 min of the constant-work rate exercise and, to evaluate the magnitude of the HR “drift” during this trial, 2-min averages around the 10th and 30th minutes were calculated.

Constructing the intensity domain schema. The exercise intensity domains (i.e., moderate, heavy, severe) were identified for each individual by determining LT and MLSS on the basis of absolute and relative $\dot{V}O_2$, work rate, and HR. Thereafter, based on the relative position of LT and MLSS, the number of individuals falling within each domain was determined at 5% intervals from 35% to 95% of $\dot{V}O_{2max}$, WR_{peak} , and HR_{max} . These intensities were chosen because they encompass the range of intensities suggested by current exercise prescription guidelines (24) and are those generally selected in research and clinical settings. All data processing and editing were performed by using customized functions of a commercially available computer software (Origin, OriginLab Corp., Northampton, MA).

Statistical Analysis

Data are presented as means $\pm$ SD. The Shapiro–Wilk test was performed to assess the normality of the distribution of each dependent variable. Comparisons between men and women in terms of maximal and submaximal indices of cardiorespiratory fitness were made using a series of independent-sample t test. Pearson’s correlation coefficients were calculated to assess the association between $\dot{V}O_{2max}$ and the indices of

exercise intensity (i.e., LT and MLSS) expressed in relative terms. To verify where the observed number of individuals distributed in a certain domain at a specific fixed percentage was significantly different compared with the number that would be expected based on a random allocation, we applied contingency tables and Pearson’s χ^2 tests. Independence of observed frequencies and expected frequencies assumptions were verified prior to the analysis. To account for any low expected frequency (i.e., $n < 5$), Fisher’s exact statistics was used. A significant χ^2 test was further broken down with standardized residual analysis, and standardized residuals were interpreted as z -scores (using $z = 1.96$ as a cutoff value to establish significance at the 0.05 level). Briefly, a standardized residual > 1.96 indicates that, at a given percentage of $\dot{V}O_{2max}$, WR_{peak} , or HR_{max} , there is a significantly larger observed number of individuals in a certain domain (i.e., *moderate*, *heavy* or *severe*) compared to a random allocation. All statistical analyses were performed using RStudio (Version 1.1.447, 2018, Boston, USA) and $\alpha = 0.05$; statistical significance was accepted when $P < \alpha$. Effect sizes [Cohen’s d , ranked as trivial (0–0.19), small (0.20–0.49), medium (0.50–0.79), and large (0.80 and greater)] (31) are also reported as objective and standardized measures of magnitude of effects and as alternative metrics of meaningfulness (32).

RESULTS

The physical characteristics of the participants and their incremental exercise results are shown in Table 1. Compared to women, men had a greater absolute ($t = 10.4$, $df = 98$, $P < 0.05$, $d = 2.1$) and relative $\dot{V}O_{2max}$ ($t = 5.2$, $df = 98$, $P < 0.05$, $d = 1.0$) and WR_{peak} (absolute: $t = 10.2$, $df = 98$, $P < 0.05$, $d = 2.0$ and relative: $t = 4.8$, $df = 98$, $P < 0.05$, $d = 0.9$), but similar HR_{max} ($t = 0.2$, $df = 98$, $P = 0.85$, $d = 0.1$).

The frequency distributions of absolute and relative $\dot{V}O_{2max}$ for men and women are displayed in Figure 1. Percent of $\dot{V}O_{2max}$ ($r = -0.191$; $P > 0.05$) and the $\%WR_{peak}$ ($r = 0.101$; $P > 0.05$) at LT were not correlated with $\dot{V}O_{2max}$ ($mL \cdot kg^{-1} \cdot min^{-1}$). There was a moderate correlation between $\dot{V}O_{2max}$ ($mL \cdot kg^{-1} \cdot min^{-1}$) and $\%WR_{peak}$ ($r = 0.419$; $P < 0.05$) but not with $\% \dot{V}O_{2max}$ ($r = 0.135$; $P > 0.05$) at MLSS.

Table 2 displays the average and range of the $\dot{V}O_2$, work rate, and HR at LT and MLSS in both absolute and relative values. Compared with men, the $\% \dot{V}O_{2max}$ and $\%HR_{max}$ at LT and MLSS were greater in women ($P < 0.05$).

The HR response during the constant-work rate trial at MLSS increased similarly in men and women progressing at the 10th and 30th minutes from 160 ± 11 bpm to 167 ± 12 bpm ($t's > 20.0$, $df = 99$, $P < 0.05$, $d's > 2.0$). The average $[La^-]_b$ at min 10th and 30th during the constant-work rate trial at MLSS was 5.1 ± 1.5 and 5.6 ± 1.6 mM, respectively.

The frequency distribution of individuals within each domain at the discrete percentages of $\dot{V}O_{2max}$, WR_{peak} , and HR_{max} are depicted in Figure 2 (panels A, B, C, D, E, and F). These data indicate that within the range of intensities typically used in research and clinical settings, in most cases, no fixed-percentage

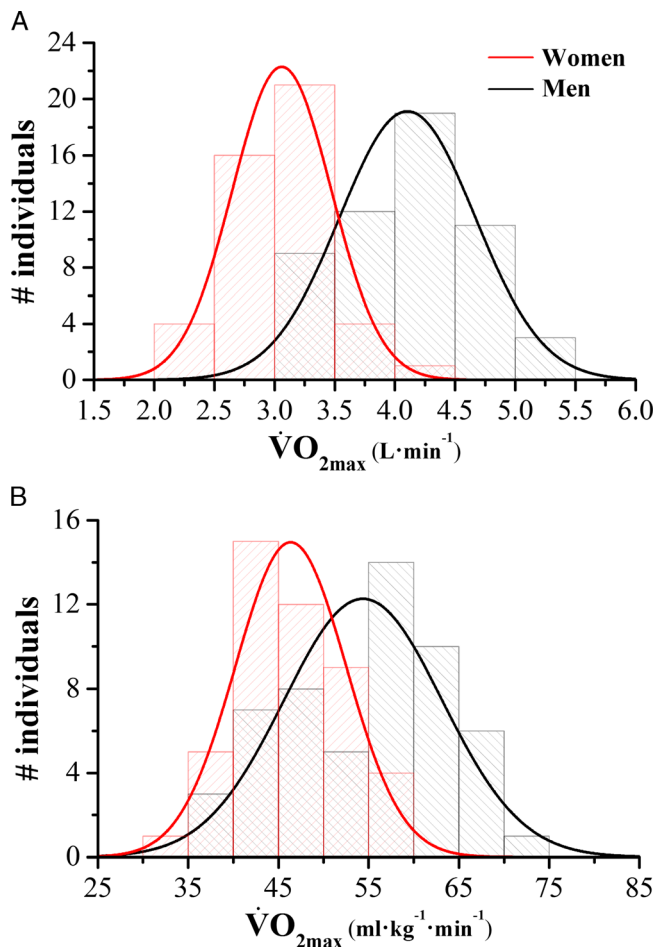


FIGURE 1—Distribution of women and men on the basis of absolute (A) and relative (B) $\dot{V}O_{2\max}$.

of maximum values could guarantee a uniform intensity domain distribution among the individuals.

Figure 3 displays χ^2 test analyses and standardized residuals calculation for women and men. There was a significant association between sexes and intensity domains at different discrete $\% \dot{V}O_{2\max}$, $\%WR_{\text{peak}}$ and $\%HR_{\max}$ (all χ^2 (24) > 1529, $P < 0.001$), meaning that individuals are classified differently based on their sex. Within the same sex, standardized residuals > ± 1.96 in Figure 3 allow us to verify at which fixed-percentage there exists a significantly different observed number of individuals within a given domain in respect to a random allocation.

DISCUSSION

In a large dataset including 100 women and men, the present study examined whether the current gold-standard exercise prescription approach, based on fixed-percentages of parameters/variables measured at maximal exercise capacity, could accurately stratify individuals within exercise-intensity domains. Findings demonstrated that, given between-subjects variability in the ranges of $\% \dot{V}O_{2\max}$, $\%WR_{\text{peak}}$, and $\%HR_{\max}$ defining each exercise-intensity domain, the fixed-percentage approach

cannot guarantee an accurate and homogeneous domain-specific exercise prescription. This carries critical implications for the interpretation of the results of acute and long-term exercise interventions and for the efficacy and applicability of current aerobic exercise prescription frameworks.

Limitations of the fixed-percentage approach to assign exercise intensity in a given domain. After the LT and MLSS were established from the ramp-incremental and constant-work rate trials, the range of $\dot{V}O_2$, work rate and HR values associated with moderate, heavy, and severe-intensity domains were identified for each individual. From these ranges, the number of participants exercising within each of these domains could be determined at any specific $\% \dot{V}O_{2\max}$, $\%WR_{\text{peak}}$, and $\%HR_{\max}$. Irrespective of the variable of choice, no single fixed-percentage (within the range of intensities typically used for exercise prescription) could guarantee a uniform domain-specific distribution during constant-work rate exercise. From Figure 2, it can be observed that within $\% \dot{V}O_{2\max}$, $\%WR_{\text{peak}}$, and $\%HR_{\max}$ ranges, individuals were distributed between two or across all three domains. This variability occurs due to the large ranges in the relative position of LT and MLSS among individuals (Table 2). Interestingly, despite the longstanding assumption that exercise intensity “thresholds” (i.e., LT and MLSS) occur at higher percentage of $\dot{V}O_{2\max}$ in individuals with a higher fitness level, we found that the relative positions of LT and MLSS were not correlated with $\dot{V}O_{2\max}$. It must be acknowledged, however, that: (i) the $\dot{V}O_{2\max}$ range of the current data set was between ~35 and ~65 $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$; had we included individuals with higher $\dot{V}O_{2\max}$, it could be possible that this correlation would have been present; and (ii) the lack of correlation might reflect the different physiological constraints that determine $\dot{V}O_{2\max}$ and the relative position of these thresholds in a heterogeneous group of individuals. Regardless, in the context of our study, these observations highlight how the relative positions of LT and MLSS—and related domains—are highly unpredictable without accurate testing.

TABLE 2. Participants' absolute and relative values of LT and MLSS on the basis of $\dot{V}O_2$, work rate, and HR.

	All (n = 100)	Women (n = 46)	Men (n = 54)
LT			
L·min ⁻¹	2.17 ± 0.43	1.90 ± 0.23 (1.50–2.50)	2.40 ± 0.43 ^a (1.60–3.30)
$\% \dot{V}O_{2\max}$	60.3 ± 6.2	62.5 ± 5.7 (49.4–73.7)	58.4 ± 6.0 ^a (45.1–72.5)
W	146 ± 38	123 ± 25 (62–175)	166 ± 37 ^a (105–250)
$\%WR_{\text{peak}}$	42.4 ± 6.6	42.0 ± 6.6 (23.4–55.3)	42.7 ± 6.7 (29.3–57.0)
bpm	137 ± 13	140 ± 14 (105–170)	135 ± 11 (114–158)
$\%HR_{\max}$	74.9 ± 5.7	76.1 ± 6.3 (59.9–89.9)	73.9 ± 5.1 ^a (63.6–82.0)
MLSS			
L·min ⁻¹	2.99 ± 0.64	2.56 ± 0.41 (1.70–3.60)	3.37 ± 0.56 ^a (2.27–4.20)
$\% \dot{V}O_{2\max}$	82.6 ± 5.4	83.4 ± 5.2 (72.6–93.8)	81.8 ± 5.4 ^a (68.5–95.8)
W	204 ± 49	174 ± 33 (104–248)	230 ± 46 ^a (135–310)
$\%WR_{\text{peak}}$	59.2 ± 5.1	59.3 ± 5.2 (47.9–70.5)	59.0 ± 5.0 (43.6–67.9)
bpm	163 ± 11	167 ± 11 (130–184.5)	160 ± 10 ^a (139–189)
$\%HR_{\max}$	89.0 ± 4.5	90.7 ± 3.9 (77.4–95.7)	87.5 ± 4.6 ^a (74.9–96.9)

Data are presented as mean ± SD (range). $\% \dot{V}O_{2\max}$: percentage of maximal rate of O_2 uptake. $\%WR_{\text{peak}}$: percentage of the highest work rate achieved at the end of the cardiopulmonary ramp-incremental test. $\%HR_{\max}$: percentage of the highest HR value recorded during the cardiopulmonary ramp-incremental test.

^aSignificance between women and men.

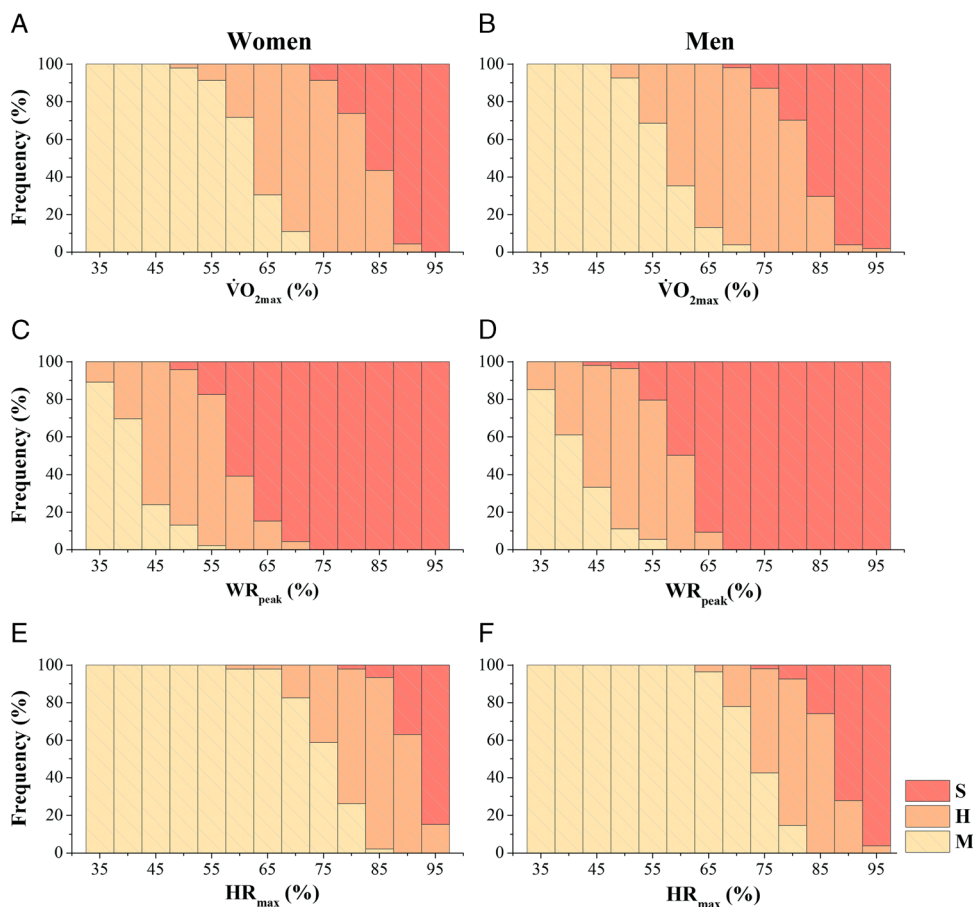


FIGURE 2—Distribution of individuals in the moderate (M), heavy (H) and severe (S) domains at discrete $\% \dot{V}O_{2max}$ (A, B), $\%WR_{peak}$ (C, D) and $\%HR_{max}$ (E, F). Women: left panels. Men: right panels.

“Critical” fixed-percentages and sex-related differences in the domain distribution. At 5% intervals of percent maximum values ranging from 35 to 95%, χ^2 analysis was used to compare the observed frequency of individuals within a given domain to that expected based on random allocation. In Figure 3, it is possible to observe significant “overlaps” across domains within the same sex and the differences in frequencies at discrete fixed-percentages between sexes. For example, “critical” fixed percentages at which significant chances of domain-overlap (i.e., moderate-heavy and heavy-severe) occurred, were: 55% to 65% and 75% to 85% of $\dot{V}O_{2max}$, 35%–55% and 60% to 65% of WR_{peak} , and 70% and 85% to 90% of HR_{max} . Regarding sex-differences, greater chances of moderate-domain at 60% and 70% of $\dot{V}O_{2max}$, and of heavy domain at 75%, 80%, and 85% of $\dot{V}O_{2max}$ were observed in women compared with men. On the basis of $\%HR_{max}$, greater chances of heavy domain at 70% of HR_{max} and of severe domain at 85% and 90% of HR_{max} were observed in men compared with women. These sex-related differences reflect the higher $\% \dot{V}O_{2max}$ and $\%HR_{max}$ at which LT and MLSS occurred in women compared with men (Table 2). Although understanding the mechanistic components underpinning these differences is beyond the scope of the present study, some established sex-related

differences in muscle fiber composition (34), oxidative capacity (35), metabolism (36), and cardiac function (37) may play a role.

Further limitations of the fixed-percent approach: the disconnection between $\dot{V}O_2$ and work rate, and HR drift. Concerning exercise prescriptions based on $\% \dot{V}O_{2max}$, the present study evaluated the distribution of domains at “true” $\% \dot{V}O_{2max}$ values (a scenario that would occur when the work rate during continuous exercise is “manually” adjusted to elicit the desired $\% \dot{V}O_{2max}$ ([18])). Particularly in research settings, however, it is a common practice to assign a work rate linearly interpolated from the $\dot{V}O_2$ –work rate relationship previously assessed during a cardiopulmonary incremental exercise test to a specific $\% \dot{V}O_{2max}$. In this context, although it could be anticipated that both approaches produce similar domain variability, an additional downfall of using an incremental-derived $\%WR_{peak}$ is that this practice will yield an unpredictable $\% \dot{V}O_2$ during constant-work-rate exercise (most likely greater than desired). The reasons for this have been thoroughly discussed in previous reviews (3,17), and relate fundamentally to the fact that the $\dot{V}O_2$ control system increasingly lags the changes in work rate during incremental exercise and that steady-state $\dot{V}O_2$ and work rate are not linearly related across all the intensity

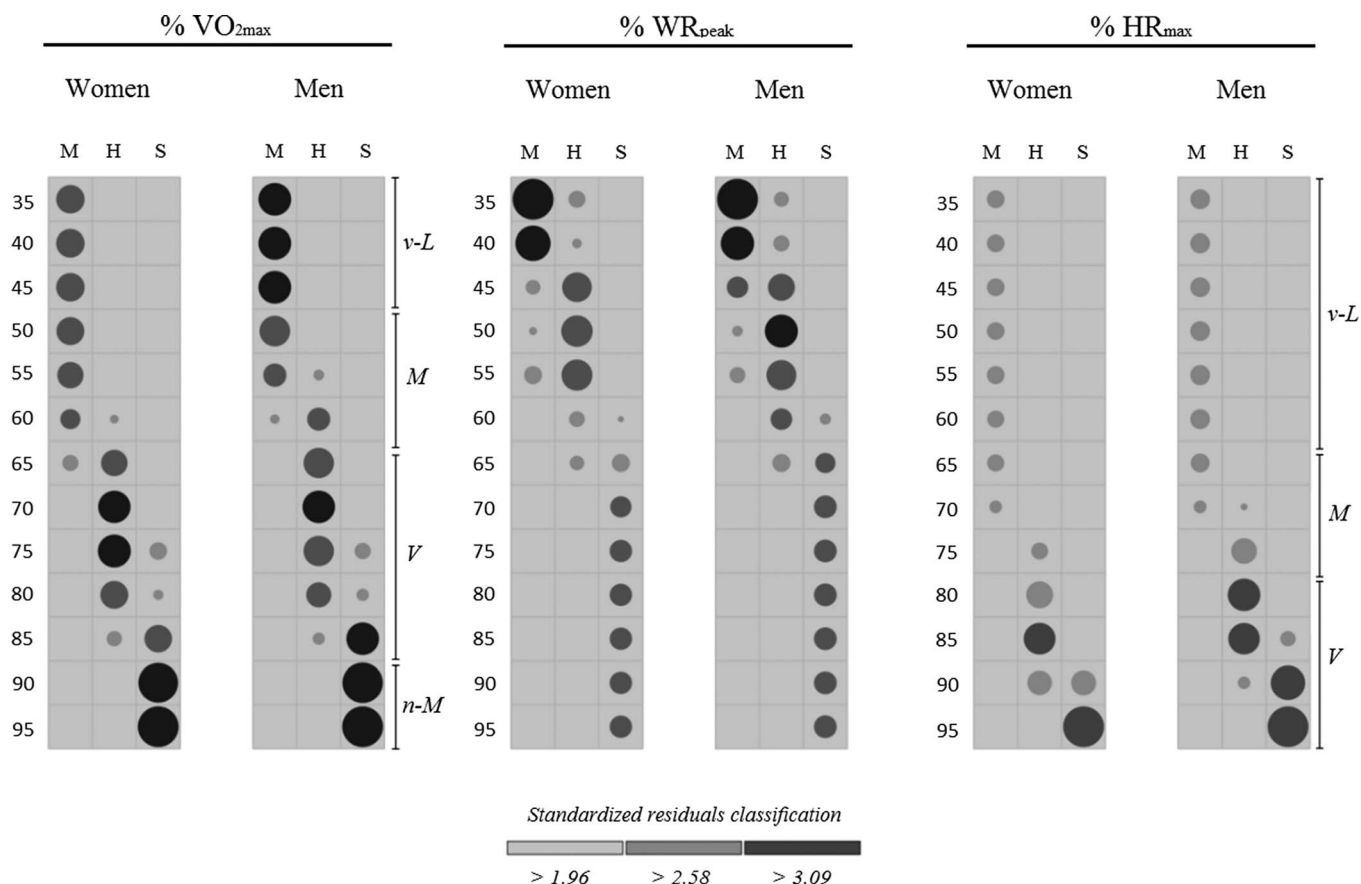


FIGURE 3—Classification plot displaying the results of the contingency tables, Pearson's χ^2 tests and standardized residuals analysis. In women and men, at discrete % $\dot{V}O_{2max}$, %WR_{peak}, and %HR_{max}, the observed and expected individual's frequencies in different exercise intensity domains (moderate [M], heavy [H], and severe [S]) were compared. Vertical lines indicate the ranges of intensities ([very-] light [v-L], moderate [M], vigorous [V], and [near]-maximal [n-M]) on the basis on % $\dot{V}O_{2max}$ and %HR_{max} identified by the ACSM's guidelines (33). A standardized residual >1.96 (in light gray) indicates a significantly larger observed number of individuals in a certain domain compared to a random allocation at a given percentage of $\dot{V}O_{2max}$, WR_{peak}, or HR_{max}. For clarity, only significant classifications are displayed. In addition, the size of the "bubble" represents the cell's contribution to the overall χ^2 statistics which can be interpreted as a measure of positive "attraction" between relative intensity (x) and domain (y).

domains (3,17). Considering these factors, exercise intensity prescription using a %WR_{peak}—even when associated to a ramp-incremental derived % $\dot{V}O_{2max}$ —should be avoided.

For exercise prescription based on fixed %HR_{max}, it is important to consider that, although a distribution of domains is presented in Figure 2, confidence in establishing exercise intensity boundaries based on HR is low. Indeed, the failure of HR to attain a steady-state response at any intensity during constant-work-rate exercise (e.g., the cardiac "drift" while exercising at MLSS was equivalent to ~4% in the present study) hinders the precise association of a univocal %HR_{max} to any exercise intensity. Additionally, in experimental settings, to ensure that HR remains at a fixed percentage of HR_{max}, the work rate would need to be progressively reduced with time, which might cause unintended transition from one domain to another and reduce the metabolic stimulus (i.e., lower $\dot{V}O_2$) (38).

Implications for exercise prescription guidelines.

The goal of current exercise prescription guidelines from the various health agencies, such as the American Heart Association and the American College of Sport Medicine, is to

promote physical activity at the population level through the adoption of "individualized exercise prescription" (24,33,39). In doing so, for instance, specific ranges of intensity [i.e., (very-) light, moderate, vigorous, (near-) maximal] based predominantly on discrete % $\dot{V}O_{2max}$ and %HR_{max} have been established (33). This framework, however, cannot account for interindividual variations in physiological thresholds and thus will not guarantee adequate control of exercise intensity. For example, within the 64% to 90% $\dot{V}O_{2max}$ and 77% to 95% HR_{max} ranges corresponding to vigorous exercise (33), individuals would be spread across the three intensity domains of exercise (i.e., moderate, heavy, and severe). Therefore, although it is recognized that exercise training can result in a wide range of health-related benefits even when using fixed-percentage approaches (40), a reevaluation of these guidelines might be needed to tailor the exercise intensity prescription (also accounting for potential sex-related differences) and optimize the health-related benefits of exercise.

Physiological implications of a "domain misclassification." When using the exercise-intensity domain approach to prescribe exercise work rates corresponding to a

specific intensity, the characteristics of the metabolic responses (e.g., $\dot{V}O_2$ and $[La^-]_b$) are highly predictable as are the rates of perceived effort and tolerable durations at any work rate (8,9,11,41). In contrast, when using fixed percentage of maximum values, the characteristics of these responses at any percent-derived value will be unpredictable. Therefore, metabolic efficiency, metabolite accumulation, fatigue etiology, and ultimately exercise tolerance at a fixed-percentage will vary greatly from individual to individual, leading to a poor control of the metabolic stress. However, despite the recognized limitations of the fixed-percentage approach (17–19,42,43), most studies, including those evaluating training responsiveness, continue to adopt this approach for exercise intensity prescription. This sheds light on the hot topic of interindividual variability in training responsiveness and the notion of exercise training “responders” versus “nonresponders” (44,45). Although genetic factors are important (21), our data highlight that the variance in physiological stress profiles from a standardized fixed-percentage prescription may play a greater role than typically attributed in determining the magnitude of the training “responsiveness.” Interestingly, a recent study showed that the incidence of nonresponders is abolished when exercise training is prescribed based on physiological “thresholds” (46). Thus, should we expect homogeneous responsiveness to training from a heterogeneous exercise intensity prescription?

Practical application of the exercise-intensity domain approach. Although the work rate at LT, and thus the confines of the moderate-intensity domain, can be accurately established from a single ramp-incremental exercise test (provided that the ramp- $\dot{V}O_2$ data are correctly “left-shifted” to account for the MRT ([47])), the accurate determination of the heavy-to-severe boundary of exercise intensity requires additional laboratory visits (generally three to five). Alternative strategies to limit the burden of additional testing have been proposed including ramp-incremental test-based equations or analysis of blood lactate profiles during sub-maximal exercise—for

MLSS estimation (25,48)—and all-out or “ramp-sprint” tests—for CP estimation (49–51). However, the viability of any of these approaches is dependent on the population being tested and the degree of accuracy required. In this context, it must be considered that any approach—including the gold standards—present some level of error (52,53). Therefore, being aware of the advantages and the limitations of each approach and knowing the physiological implications of exercising in the heavy *versus* in the severe domain can help identifying the best strategy to determine in any context and with accuracy the heavy-to-severe boundary of exercise intensity.

CONCLUSIONS

Exercise prescription based on fixed percentages of maximum provides an inaccurate means for controlling exercise intensity. Given that accurate characterization of the training stimulus is critical to obtain the desired metabolic stimulus and subsequent adaptations to exercise training, a model that considers the exercise intensity domains for exercise prescription is recommended. Application of this approach would optimize health-related outcomes of participants and better characterize the molecular and system-level adaptations related to acute and chronic exercise trainings. Current exercise prescription guidelines that propose percent-based ranges of maximum cannot guarantee an appropriate individualization of the intensity prescription, and this should be carefully considered when presenting exercise and physical activity guidelines.

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